

SURAJ SHIVAKUMAR

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Education

University of Washington - Seattle

Sep. 2025 – June 2028 (Estimated)

Bachelor's of Science in Electrical and Computer Engineering - GPA: 3.9/4.0

Seattle, WA

Skills

Technical: Java, Python, C++, KiCad, STM32CubeIDE, Raspberry Pi, Arduino, Git/GitHub, Microsoft 365, Figma

Interpersonal: Team Leadership, Attention to Detail, Outgoing, Organization, Communication

Experience

Research and Development Lead

October 2025 – Present

Husky Flying Club: Autonomous Flight Team's CSE Division

University of Washington

- Managing a 5-person team delivering a custom STM32-based flight controller PCB, Power Distribution Hub (PDH), and Electronic Speed Controller (ESC) stack for autonomous quadcopters.
- Designing and reviewing embedded schematics and multilayer PCBs in KiCad, following STM32 best practices for clocking, reset/boot, SWD/SWO debugging, and high-speed routing to ensure reliable hardware performance.
- Configuring and integrating firmware-aware hardware in STM32CubeIDE, including pin multiplexing, clock-tree setup, peripheral enablement, and debug interface alignment, enabling smooth software-hardware integration.
- Engineering robust power delivery, sensing, and protection systems with 3.3V/5V regulation, decoupling, rail separation, filtering, and EMI-aware layouts to ensure stable and safe operation.
- Integrating and validating flight-critical interfaces (IMU, GPS, magnetometer, barometer, CAN, PWM, UART, I²C, SPI, USB) to enable accurate sensor fusion and autonomous flight control.

Nonprofit Founder, Board Member

2023 – Present

Hackabyte — hackabyte.org

Redmond, WA

- Collaborating with a team of 6 to organize two-day, in-person/online hackathons with 1,000+ total teen participants.
- Secured sponsorships from leading companies (AoPS, Desmos, Costco), fundraising \$15,000+ for operations/growth.
- Leading a team of 20+ organizers and volunteers to design and execute efficient, data-driven marketing strategies.
- Coordinating technical workshops led by industry professionals and judges from major technology companies.
- Founded subsidiary program, LUMA, empowering 500+ students to learn coding through hands-on classes.
- Expanded hackathon and educational events into California and Texas, increasing organizational reach and impact.

Projects

Genre Identifier | Python (Librosa, NumPy), PyTorch, TensorFlow, FastAPI

December 2025

- Built an audio genre classification system using DSP and deep learning to identify music genres from raw audio files.
- Developed an audio preprocessing pipeline with Librosa, NumPy, and SciPy, including band-pass filtering, harmonic-percussive source separation (HPSS), mel-spectrogram/CQT feature extraction, and audio normalization.
- Implemented and trained a CRNN (CNN + LSTM) to learn time-frequency patterns and genre temporal structure.
- Deployed the model with a FastAPI/Flask backend enabling audio upload and real-time genre prediction via web UI.

AI-Driven WiFi Optimization Tool | Python, Flask, Raspberry Pi, WebSockets

October 2025

- Developed an intelligent router placement system with Python (Flask-SocketIO) and Raspberry Pi, converting 5G telemetry into heatmaps and AI-driven recommendations.
- Built a Flask REST API with thread-safe data, GPS mapping, and calibration, streaming metrics to visualizations.
- Integrated Google Gemini AI for placement guidance, speed analytics, and predictive WiFi insights via WebSocket.

Shipso: An End-to-End Shipping Platform | React, Node.js, Figma, SQL

Present

- Building a platform to centralize communication, manage data, and workflow automation for shipping stakeholders.
- Developing role-specific dashboards with AI-predicted tasks, progress tracking, and automated notifications using Figma.

Additional Work Experience

Math Teacher/Tutor

2020 – Present

Anagha's Math — anaghasmath.com

Part Time – Hybrid

- Teaching 30+ students from 1st grade through Calculus BC (college level) concepts for school preparation.
- Guided 100+ students as a teaching assistant, answering questions and advising on homework packets.
- Designed test-prep and summer course curricula, contributing to 500+ hours of combined volunteer and paid work.